

High Electricity Bills Due To Inefficient Household Appliances

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Abstract—The high use of electricity in homes is mainly due to inefficient appliances, lack of user awareness, and the absence of real-time energy monitoring. Traditional electricity bills only show monthly usage, making it difficult for homeowners to identify which appliances waste the most energy, including power used in standby mode. This study discusses how Internet of Things(IoT)-based Smart Energy Monitoring Systems(SEMS) can help solve this problem by providing real-time and appliance level energy data. It reviews existing research, commercial products, and open-source solutions, focusing on common monitoring methods such as whole-home monitoring, individual appliance monitoring, and Non-Intrusive Load Monitoring (NILM). The study also explains the importance of communication technologies, clear data display, and simple analysis methods that provide useful feedback to users. Previous research shows that when people can clearly see their energy use, they are more likely to change their habits and save electricity. The study highlights the need for affordable, easy-to-use, and scalable energy monitoring systems for households, which can help reduce electricity bills and support more sustainable energy use.

Index Terms—Smart Energy Monitoring, Household Appliances, Real-Time Energy Monitoring, Internet of Things, Energy Efficiency

I. INTRODUCTION

In many homes, high electricity bills are a direct result of inefficient appliance usage, with most consumers lacking the specific data needed to identify the main culprits in energy. This lack of detailed visibility makes it difficult to pinpoint waste, whether from outdated appliances, standby power, or poor habits. To solve this, a Smart Energy Monitoring System (SEMS) will use IoT technology to provide homeowners with real-time, appliance-specific energy consumption data. This information, delivered through an intrusive interface, empowers users to make informed decisions and take action to reduce energy usage, ultimately lowering their electricity bills.

II. LITERATURE REVIEW

High energy bills are a common problem in many households and are often the result of insufficient appliance usage. This issue is frequently obscured by aggregated monthly electricity bills, which do not reveal which specific devices are responsible for excessive energy consumption. A considerable share of this hidden waste originates from so-called "phantom

loads," referring to electricity consumed by appliances left in standby mode. A study published in Nature Energy in 2018 by Tiefenbeck et al. demonstrated that providing users with real-time energy consumption feedback can significantly influence energy-saving behavior, even in the absence of direct financial incentives. This finding highlights the importance of delivering detailed and appliance-level energy information to promote more efficient household energy use.

Advancements in Internet of Things (IoT) technology have transformed the way energy consumption is monitored within residential environments. The emergence of Smart Energy Monitoring Systems (SEMS) enables real-time tracking of electricity usage and provides insights that were previously unavailable to consumers. Research identifies two primary sensing approaches: whole-home monitoring and individual appliance monitoring. Whole-home systems utilize sensors installed at the main electrical panel to measure overall and circuit-level energy consumption, while advanced methods such as Non-Intrusive Load Monitoring (NILM) analyze aggregate electrical signatures to estimate the consumption of individual appliances without requiring multiple sensors.

In whole-home monitoring solutions, devices such as the Emporia Vue Smart Energy Monitor employ current transformers to deliver a detailed breakdown of energy usage across different circuits. This approach offers homeowners a comprehensive overview of household electricity consumption and helps identify high-energy-consuming circuits. Emporia's system exemplifies how centralized monitoring can support informed energy management decisions by presenting clear and structured consumption data.

Individual appliance monitoring focuses on directly measuring the energy usage of specific devices through smart plugs or smart outlets. Commercial products such as the TPLINK Tapo P110M integrate energy measurement capabilities, allowing users to track consumption at the appliance level. In addition to these commercial solutions, research has increasingly explored low-cost, do-it-yourself alternatives that utilize affordable hardware components, including ESP32 micro-controllers and PZEM-004T sensors. These DIY systems demonstrate the feasibility of creating accessible and cost-effective energy monitoring solutions, expanding the reach of smart energy technologies.

Communication protocols are a critical element of IoT-based energy monitoring systems, influencing factors such as power consumption, data reliability, and system scalability. Wi-Fi remains a popular choice due to its high data throughput and widespread availability in households. However, low-power protocols such as Zigbee are also widely adopted, particularly because of their mesh networking capabilities, which enhance coverage and reliability. Many manufacturers, including Aqara, rely on Zigbee-based communication for smart energy devices. Additionally, open-source firmware solutions such as Tasmota enable users to reconfigure compatible smart plugs for local monitoring and control, reducing the dependence on proprietary cloud platforms.

The effectiveness of a Smart Energy Monitoring System ultimately depends on its ability to analyze collected data and present it in a meaningful and actionable format. Research indicates that advanced data analytics, including machine learning techniques, can reveal hidden consumption patterns and support energy demand forecasting. A meta-analysis conducted by Karlin et al. in 2015 confirmed that timely and well-designed user feedback plays a crucial role in driving energy conservation behavior. Modern SEMS platforms employ intuitive mobile applications and web-based dashboards to visualize energy usage trends and provide actionable recommendations. Platforms such as ThingsBoard and Home Assistant further enhance system functionality by enabling customization, automation, and integration with multiple devices, thereby empowering homeowners to reduce energy waste and manage electricity consumption more effectively.

III. PROBLEM STATEMENT

Homeowners face high electricity bills due to inefficient household appliance usage, often without awareness of which specific devices are the biggest energy consumers. The absence of a real-time, appliance-specific energy monitoring solution prevents households from identifying energy waste and making informed decisions to reduce their overall consumption. This leads to unnecessary financial costs and increased environmental impact from energy overuse.

IV. SYSTEM OVERVIEW

The proposed system is an IoT-based Smart Energy Monitoring System developed to monitor, analyze, and manage electrical energy consumption in a residential environment. The main objective of the system is to provide appliance-level energy monitoring, calculate energy usage and electricity cost, and generate alerts when energy consumption exceeds predefined thresholds.

Unlike conventional energy meters that provide only total consumption, the proposed system enables individual appliance monitoring allowing users to identify high energy consuming devices such as water heaters, dishwashers and standby loads. This improves user awareness and supports energy-efficient decision-making.

The system operates by measuring electrical parameters, computing instantaneous power, integrating power to obtain

energy, and estimating cost based on a predefined tariff. A threshold-based alert mechanism continuously monitors consumption and indicates abnormal usage conditions.

The functional behavior of the system is modeled and validated using MATLAB/Simulink. The Simulink model includes appliance-level monitoring subsystems, power and energy computation blocks, threshold detection logic, and a simulated MQTT-based IoT communication layer. This communication layer represents real-time data transmission of power, energy, cost, and alert signals from the edge device to a cloud-level dashboard, enabling remote monitoring and visualization consistent with modern smart grid and IoT architectures.

In addition to the computational modeling, the physical realization of the system is addressed through mechanical and hardware prototyping using SolidWorks. A compact enclosure is designed to house the custom PCB, ESP32 micro-controller, relay modules, power supply unit, and wiring interfaces. The SolidWorks assembly ensures proper component placement, electrical isolation, ventilation considerations, and ease of installation in residential settings. This electromechanical integration bridges the gap between simulation and real-world deployment demonstrating the feasibility of the proposed system beyond software validation.

Overall, the system consists of four main components:

1. Appliance-level monitoring units,
2. Power, energy, and cost calculation module,
3. Threshold detection and alert system, and
4. IoT communication and visualization layer,

supported by a mechanically robust and compact enclosure design. The combined MATLAB/Simulink modeling and SolidWorks-based mechanical design provide a comprehensive framework for validating both the functional performance and physical implementation of the proposed Smart Energy Monitoring System.

V. SYSTEM DESIGN

The system is designed in a modular and scalable manner using MATLAB/Simulink. Each functional block represents a real-world component of a smart energy monitoring system.

A. Appliance-Level Monitoring

Each household appliance is modeled as an independent subsystem to represent its rated power consumption and operational behavior. Appliances such as refrigerators, water heaters, dishwashers, lighting loads, televisions, and standby loads are included in the model.

This modular approach allows:

- Independent monitoring of each appliance
- Easy addition or removal of appliances
- Accurate representation of different power ratings

Appliance outputs are combined to obtain the total household power consumption, which serves as the primary input for energy calculation and alert detection.

B. Power, Energy, and Cost Calculation

Instantaneous power is calculated using the product of voltage and current signals. This power signal represents real-time energy usage and varies according to appliance operation.

Energy consumption is obtained by integrating the instantaneous power over time. This ensures that energy values increase monotonically, reflecting realistic electricity consumption behavior.

The total electricity cost is calculated using the accumulated energy and a predefined tariff rate. This enables users to observe not only energy usage but also its financial impact, which is essential for practical energy management.

C. Threshold and Alert Mechanism

To detect abnormal or excessive energy usage, threshold values are defined for individual appliances and total power consumption. Each measured signal is compared with its corresponding threshold using comparison logic.

When a threshold is exceeded, an alert is generated. Individual appliance alerts are combined using a logical OR operation to produce a global alert signal. This ensures that the system raises an alert even if a single appliance causes excessive consumption.

Visual indicators are used to represent system status:

- Green indicates normal operation
- Red indicates threshold violation or abnormal usage

This design provides an intuitive and user-friendly alert mechanism suitable for real-time monitoring.

D. IoT Communication Design Using MQTT

An IoT communication layer is implemented using a simulated MQTT publish-subscribe architecture. The calculated power, energy, cost, and alert signals are packaged together and transmitted at fixed intervals.

A Zero-Order Hold (ZOH) block is used to model the MQTT publish rate, ensuring that data is sent periodically rather than continuously. This data is combined using a multiplexer to form a single payload, representing a MQTT message.

On the subscriber side, the transmitted data is decoded and visualized using displays and indicator lamps, simulating a cloud-based dashboard or mobile application. This approach accurately represents real-world IoT-based energy monitoring systems while avoiding the complexity of real network connections.

VI. MATLAB MODELS AND SIMULATION RESULTS

A. MATLAB/Simulink Model Description

The complete Smart Energy Monitoring System is modeled using MATLAB/Simulink. The model includes appliance subsystems, power and energy calculation blocks, threshold comparison logic, data logging components, and an IoT MQTT gateway.

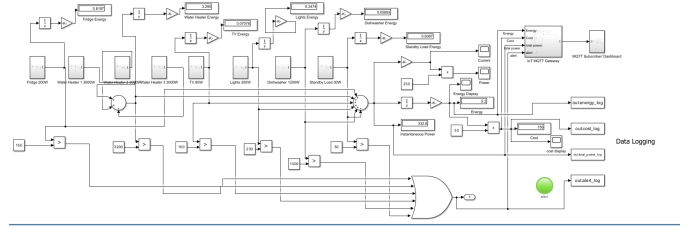


Fig. 1. Overall Simulink model of the IoT-based Smart Energy Monitoring System

B. Simulation Results

Simulation results validate the correct operation of the proposed system. The instantaneous power waveform clearly reflects changes in appliance operation, confirming accurate real-time monitoring. Energy consumption increases smoothly over time, demonstrating correct integration of power signals.

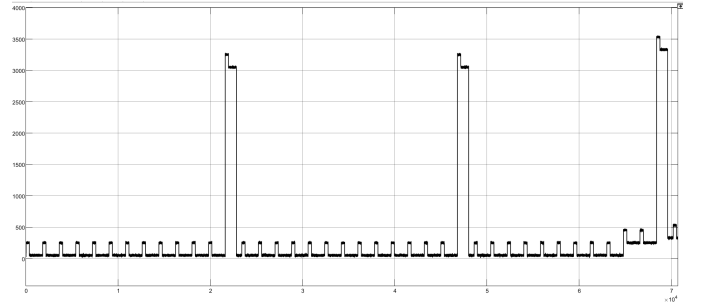


Fig. 2. Instantaneous power variation with appliance operation

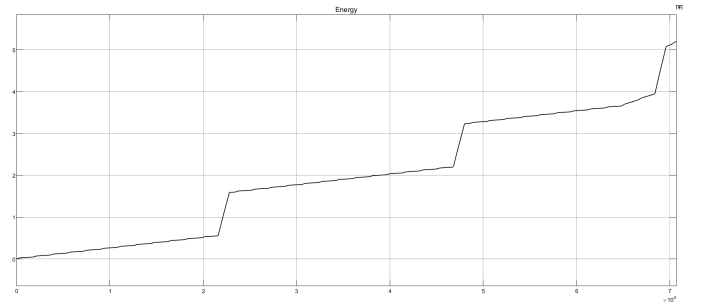


Fig. 3. Accumulated energy consumption over time

When predefined thresholds are exceeded, the alert mechanism activates correctly, and the alert indicator changes state accordingly. The alert behavior confirms the effectiveness of the threshold-based monitoring system.

The IoT MQTT simulation successfully transmits power, energy, cost, and alert data to the subscriber dashboard at fixed intervals. The subscriber displays correct numerical values and visual indicators, confirming proper data packaging, transmission, and decoding.

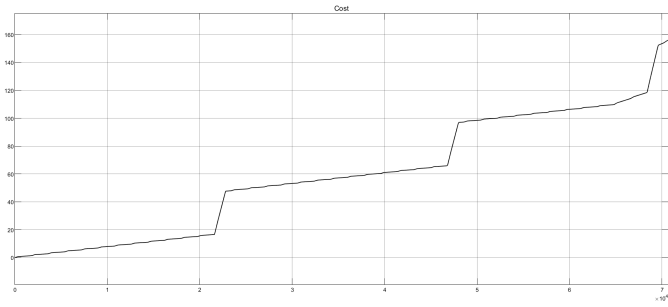


Fig. 4. Estimated electricity cost based on energy consumption

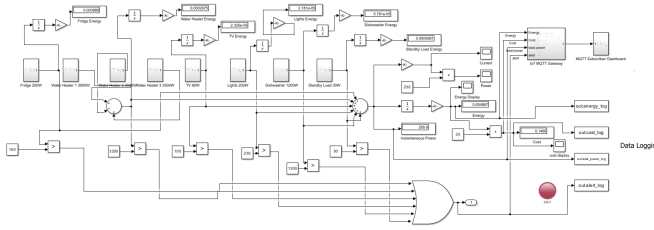


Fig. 5. Threshold-based alert activation during high power usage

C. Observations and Discussion

The simulation demonstrates that the proposed system can effectively:

- Monitor appliance-wise energy consumption
- Calculate real-time power, energy, and cost
- Detect abnormal usage conditions
- Transmit data using an IoT-based communication model

The results confirm that the system is suitable for residential smart energy monitoring applications and can be extended for real-time deployment using actual sensors and communication hardware.

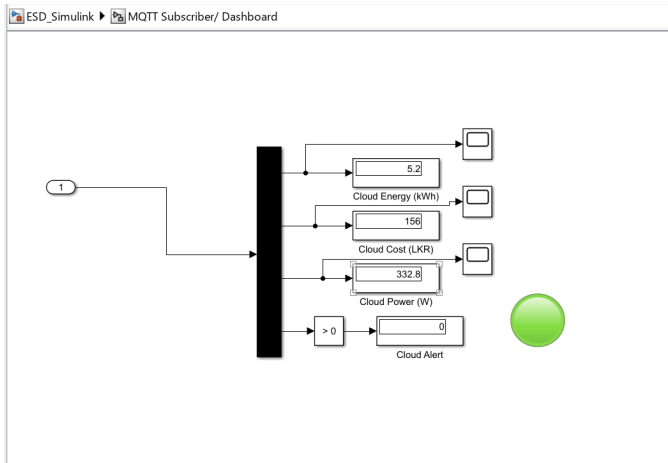


Fig. 6. MQTT subscriber dashboard displaying transmitted energy data

VII. HARDWARE SYSTEM DESIGN

A. Schematic and Circuit Protection

The control logic is built around the ESP32-DevKitC. To ensure reliability when driving inductive loads (relays), the circuit incorporates a specific protection stage. As seen in the Figure 7, each relay coil is driven by a 2N2222 transistor and protected by a 1N4007 flyback diode. This prevents high-voltage back-EMF spikes from damaging the micro-controller.

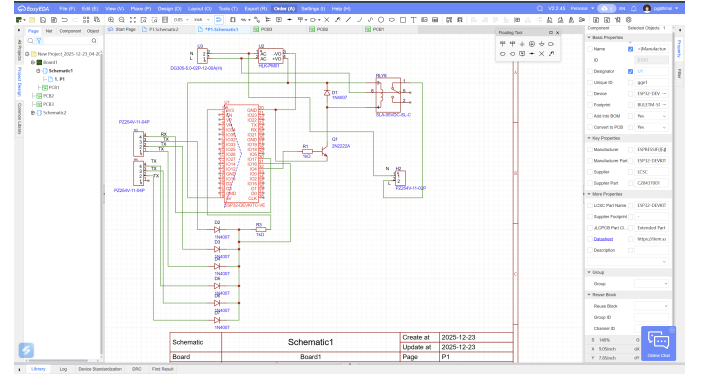


Fig. 7. Schematic and Circuit Protection

B. PCB Layout

A custom two-layer PCB was designed to accommodate the ESP32, the HLK-PM01 AC-DC power supply, and the relay banks. The layout (see Figure 8) prioritizes trace width for high-current paths and maintains isolation gaps between the AC mains and the low-voltage DC logic to meet safety standards.

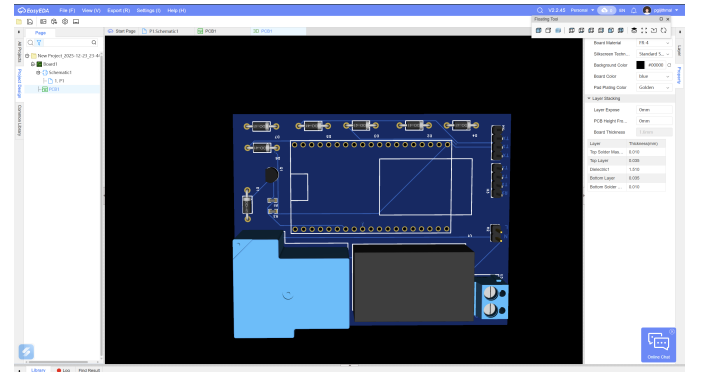


Fig. 8. PCB Layout

VIII. MECHANICAL ENCLOSURE DESIGN

A. Assembly Overview

The enclosure was modeled in SolidWorks to simulate a realistic installation scenario. Figure 9 displays the complete assembly, including the external electrical loads (Immersion Heater and CFL bulbs). The 3D modeling of the wiring harnesses ensures that the enclosure glands are sized correctly for the specific cable diameters used.

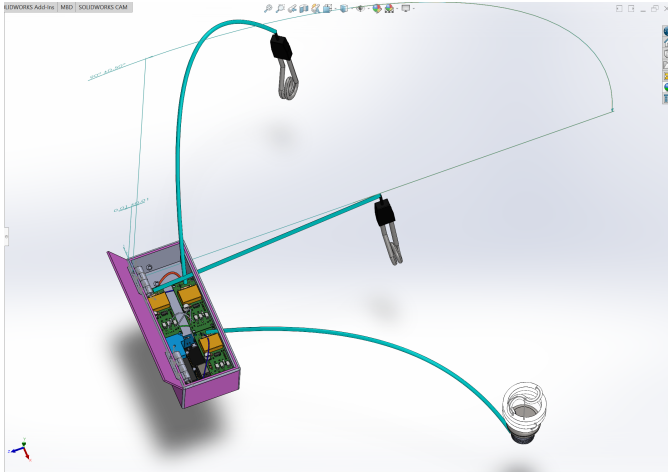


Fig. 9. Assembly Overview

B. Internal Configuration

The internal arrangement (shown in Figure 10) is critical for airflow and heat management. The PCB is mounted on standoffs to allow air circulation beneath the board. The transparent view highlights the spatial constraints, showing the tight integration of the relays and the power supply module within the ABS plastic housing.

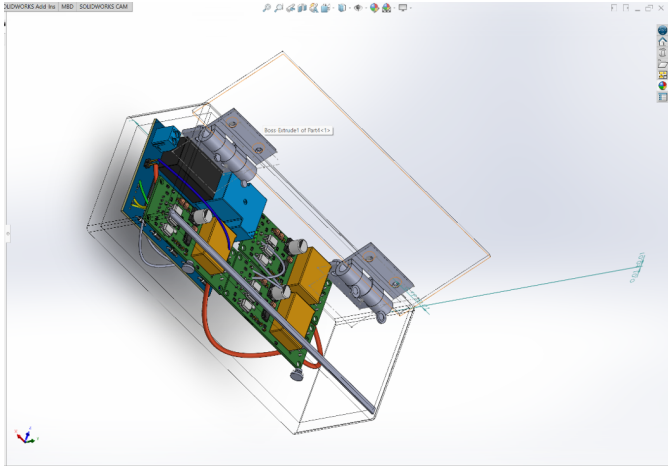


Fig. 10. Internal Configuration

IX. THERMAL SIMULATION METHODOLOGY

To validate the design before physical prototyping, a thermal model was established using SolidWorks Flow Simulation.

A. Computational Domain

The simulation methodology utilizes an "External" analysis type to model natural convection around the device. Figure 11 illustrates the computational domain (the wireframe box) and the boundary conditions. The gravitational vector is aligned with the Y-axis to correctly simulate buoyancy-driven airflow (heat rising).

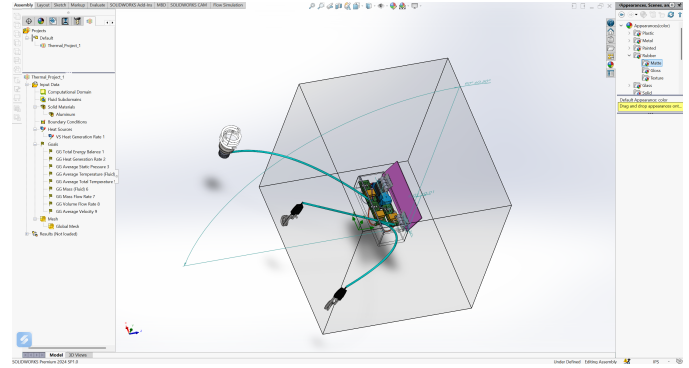


Fig. 11. Computational Domain

B. Heat Source Definition

For the upcoming thermal analysis, volumetric heat sources are assigned to the internal components based on their power dissipation.

- Relays : 0.5W (Coil excitation).
- Power Supply : 1.5W (Conversion inefficiency).
- ESP32 : 0.8W (Active WiFi transmission).

The external loads (visible in Figure 9) are excluded from the thermal calculation as they dissipate heat to the ambient environment, not the enclosure interior.

X. THERMAL SIMULATION RESULTS AND ANALYSIS

The SolidWorks Flow Simulation generated three distinct visualization plots: Cut Plots, Surface Plots, and Flow Trajectories. These visual datasets provide critical insights into the thermal reliability of the Smart Energy Monitoring System (SEMS).

A. Fluid Temperature Distribution (Cut Plot)

Figure 12 illustrates a 2D Cut Plot slicing through the center of the enclosure. This graph visualizes the ambient air temperature distribution surrounding the electronic components.

- Interpretation: The plot reveals a stable internal ambient temperature ranging between 20°C (dark blue) and 24°C (light blue/green). The absence of high temperature "red zones" in the fluid volume indicates that the enclosure volume is sufficient to dissipate the heat generated by the internal components without creating dangerous heat pockets. This validates the enclosure sizing strategy.

B. Component Thermal Stress (Surface Plot)

While the Cut Plot shows air temperature, the Surface Plot (Figure 13) maps the temperature directly onto the solid faces of the PCB and components.

- Interpretation: The simulation identifies the ESP32 microcontroller as the primary heat source, indicated by the red thermal signature (approx. 28°C). This localized heating is expected due to the active WiFi transmission power consumption (0.8W). However, the surrounding PCB area remains in the cool blue spectrum (22°C), confirming that the PCB layout effectively acts as a heatsink,

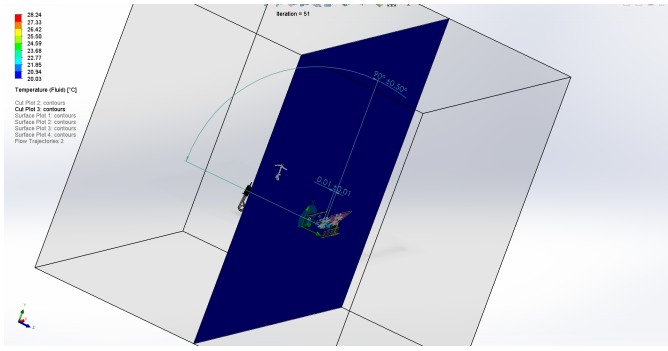


Fig. 12. Fluid Temperature Distribution (Cut Plot)

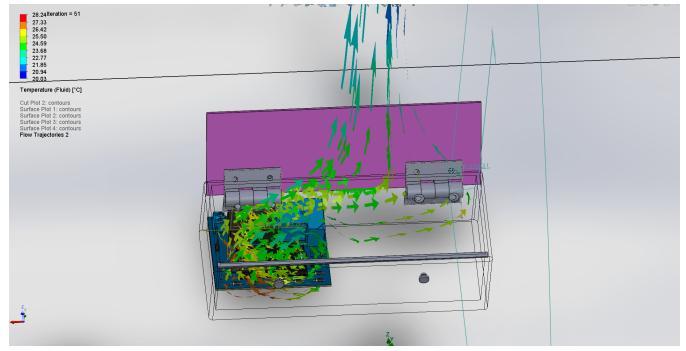


Fig. 14. Natural Convection Path (Flow Trajectories)

spreading the thermal load away from sensitive silicon components.

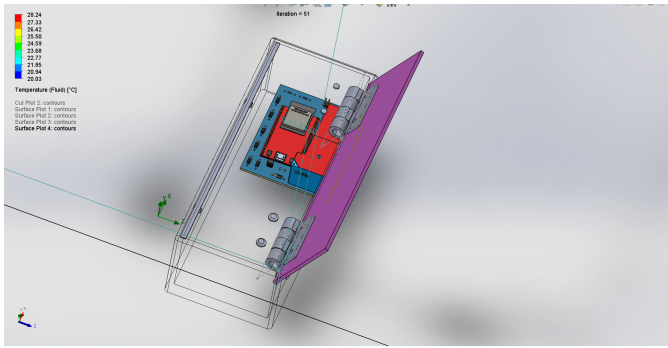


Fig. 13. Component Thermal Stress (Surface Plot)

C. Natural Convection Path (Flow Trajectories)

Figure 14 visualizes the movement of air within the sealed box using 3D Flow Trajectories. The arrows indicate the direction of velocity, while the color represents temperature.

- Interpretation: The trajectories demonstrate a clear pattern of natural convection. As air near the ESP32 and power supply warms up, it becomes less dense and rises toward the top lid of the enclosure (indicated by the upward-moving green arrows). As this air cools against the plastic walls, it cycles back down. This proves that even without an active cooling fan, the natural circulation of air is sufficient to maintain operational temperatures well below the component safety limits (typically 85°C for consumer electronics).

D. Product Reliability and Usefulness

The data derived from these simulations serves as key evidence for the product's viability in real-world scenarios:

- 1) Continuous Operation: The low maximum temperature (28°C) confirms the device can operate 24/7 without thermal throttling or degradation.
- 2) Safety Assurance: The simulation proves there is zero risk of the plastic enclosure melting or deforming, as

the internal air temperature remains well below the ABS glass transition temperature (105°C).

- 3) Sealed Design Validation: The results confirm that a fully sealed (IP-rated) enclosure is feasible, allowing the product to be used in dusty or humid environments without requiring ventilation holes that would compromise protection.

XI. CONCLUSION

The proposed IoT-based Smart Energy Monitoring System effectively addresses the problem of high household electricity bills by providing real-time, appliance-level energy monitoring, cost estimation, and threshold-based alerts. MATLAB/Simulink simulations confirmed accurate power and energy computation, reliable alert generation, and successful data transmission using a MQTT-based IoT communication model. The inclusion of hardware schematic design, PCB layout, and SolidWorks-based enclosure modeling further demonstrates the feasibility of practical implementation with appropriate electrical protection, compact integration, and thermal considerations. Overall, the system enhances user awareness of energy consumption patterns, supports informed decision-making, and offers a scalable and cost-effective solution for improving residential energy efficiency and reducing unnecessary energy waste.

XII. INDIVIDUAL CONTRIBUTION

The work presented in this project was carried out through a collaborative effort among all group members, with clearly defined responsibilities to ensure effective development and documentation of the system.

P.H.N.D.Jayawardana (230296P) was responsible for conducting the literature review, organizing the technical content, and preparing the research paper using LaTeX in accordance with the IEEE conference formatting guidelines.

P.G.I.Jithmal (230306B) focused on the mechanical and hardware prototyping aspects of the project, designing the enclosure and system layout using SolidWorks to represent the physical realization of the proposed system.

T.M.Kannangara (230314X) was primarily responsible for the design, implementation, and simulation of the Smart

Energy Monitoring System using MATLAB/Simulink, including appliance-level modeling, power and energy calculations, threshold detection, and IoT communication logic.

All members contributed to discussions, system integration, and review of the final submission.

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